Supplemental Material

S1 Simulations of hard-core gas

Hard-core gas simulations were performed in the software package DynamO [1] in arbitrary units (arb. u.) of mass $\mathcal{M}$, time $\mathcal{T}$, length $\mathcal{L}$, and energy $\mathcal{E} = \mathcal{M}\mathcal{L}^2/\mathcal{T}^2$, with $N = 4 \times 10^3$ particles of diameter $\sigma = 0.02\mathcal{L}$ and mass $\mathcal{M}$ in a $\mathcal{L} \times 0.4\ \mathcal{L} \times 0.4\ \mathcal{L}$ box. Periodic boundary conditions were applied along y and z axes, and two thermalizing reflective walls, at the temperatures $T_{\rm left}$ and $T_{\rm right}$, confined the gas between $x = 0$ and $x = \mathcal{L}$, respectively. The exact parameters of the walls' thickness and positions were chosen so as to allow for the centers of finite-size particles to access positions within $x \in (0, \mathcal{L})$ up to the machine-precision error.

The initial state of the simulation was prepared with the particles placed at the nodes of a regular rectangular lattice and their velocities sampled from the Maxwell distribution at the temperature $T = 2/(T_{\rm left}^{-1} + T_{\rm right}^{-1})$ (the harmonic mean temperature of the walls) with $k_{\rm B} = 1$. The system was then allowed to relax for 10^7 events (collisions).

To construct an effective lattice model of a hard-core gas, we consider the projection of the states onto the domain $x \in (0, L\Delta x)$ partitioned into L bins of size $\Delta x = \mathcal{L}/L$. We define the coarse-grained model of energy levels by choosing a time scale Δt and a grid $\mathcal{H}_{i=1,\dots M} = \varepsilon i^2/2$, $\mathcal{H}_M = \infty$, where $\varepsilon = \mathcal{M}(\Delta x/\Delta t)^2$.

By modeling the probability of a transition between the energy states (ℓi) and (mj) as

$$P_{\ell i|mj}(\beta_{\rm ref}) = \frac{\hat{w}_{\ell i|mj} \exp\big(-\beta_\ell(\epsilon_i - \epsilon_j)\big)}{\sum_{\ell' i'} \hat{w}_{\ell' i'|mj} \exp\big(-\beta_{\ell'}(\epsilon_{i'} - \epsilon_j\big)}\bigg|_{\beta_\ell = \beta_{\ell'} \equiv \beta_{\rm ref}} = \frac{\hat{w}_{\ell i|mj} \exp\big(-\beta_{\rm ref}\epsilon_i\big)}{\sum_{\ell' i'} \hat{w}_{\ell' i'|mj} \exp\big(-\beta_{\rm ref}\epsilon_{i'}\big)}, \tag{S1}$$

we can extract the trajectory weights $\hat{w}_{\ell i|mj}$ from statistics of transitions in equilibrium simulations at $\beta_{\rm ref} < [k_{\rm B}\max(T_{\rm left}, T_{\rm right})]^{-1}$. In particular, we collect transition numbers $n_{\ell i|mj}(f = 1, 2, \mathcal{F})$ from $\mathcal{F}+1$ consecutive frames $f = 0, \dots \mathcal{F}$. Then the probability of a transition $(\ell i|mj)$ is estimated by

$$P_{\ell i|mj} \approx \frac{\sum_{f=1}^{\mathcal{F}} n_{\ell i|mj}(f)}{\sum_{\ell=1,i=1,f'=1}^{L,L,\mathcal{F}} n_{\ell i|mj}(f')}.$$

Using these estimates we solve Eq. (S1) for $\hat{w}_{\ell i|mj}$ subject to a normalization condition

$$\sum_{\ell i} \hat{w}_{\ell i|mj} = 1$$

for all pairs (mj), and obtain

$$\hat{w}_{\ell i|mj} = \frac{P_{\ell i|mj} \exp(\beta_{\rm ref}\epsilon_i)}{\sum_{\ell' i'} P_{\ell' i'|mj} \exp(\beta_{\rm ref}\epsilon_{i'})}.$$

By a similar token we can estimate normalized degeneracies $\hat{g}_{\ell i}$ from the frequencies

$$p_{\ell i} \simeq \frac{\sum_{f=1}^{F} n_{\ell i}(f)}{N\mathcal{F}},$$

where $n_{\ell i}(f)$ is the number of particles in the state (ℓi) in the snapshot f. Then, under the normalizing condition $\sum_{\ell i} \hat{g}_{\ell i} = 1$ we have

$$\hat{g}_{\ell i} = \frac{p_{\ell i} \exp(\beta_{\rm ref}\epsilon_i)}{\sum_{\ell' i'} p_{\ell' i'} \exp(\beta_{\rm ref}\epsilon_{i'})}.$$

Parameter	**Value**
Particle diameter σ	0.366 nm
Particle mass $\mathcal{M}$	6.63×10^{-26} kg
Length constant $\mathcal{L}$	18.3 nm
Temperature T_{ref}	300 K
Energy constant $\mathcal{E}$	4.14×10^{-21} J
Time constant $\mathcal{T}$	73.2 ps
Time step Δt	2.9 ps
Bin size $\mathcal{T}$	0.732 nm
Temperature of the left wall T_{left}	273 K
Temperature of the right wall T_{right}	158 K
Temperature drop ΔT	-115 K

Table S1: Approximate correspondence between the parameters of the hard-core gas and argon at $T_{\text{ref}} = \mathcal{E}/k_{\text{B}}$ [2]

Note that at lower β particles explore higher energy states and, correspondingly, higher velocities. Therefore, to obtain an accurate match between the atomistic model of a hard core gas and its representation by the Boltzmann gas, large temperatures $T_{\text{left,right}}$ (lower beta β_{ref}) require fine resolution in space and sufficient statistics of observations for transitions involving high levels ϵ_i.

In practice for $\beta \geq 1/(k_{\text{B}}\mathcal{E})$ we obtain a Boltzmann gas model for $\beta_{\text{ref}} \leq \beta_{\text{left}} \leq \beta_{\text{right}}$ already with $L = 25$ bins, and $M = 5$ levels, and $\mathcal{F} = 10^4$ transitions (Fig. 2). Specifically, we calibrate the parameters $\hat{w}_{\ell i}$ and $\hat{g}_{\ell i}$ from molecular-dynamics simulations at $\beta_{\text{ref}} = 1/(k_{\text{B}}\mathcal{E})$, which we run for a duration of $400\mathcal{T}$ collecting snapshots at intervals $\Delta t = 0.04\mathcal{T}$ ($\varepsilon = \mathcal{E}$). In addition we retain only energy levels which register at least 100 outbound transitions.

Estimation step for the degeneracies $g_{\ell i}$ ensures that the equilibrium distribution of mass is consistent with the Boltzmann law in the coarse-grained model near the wall ends. Whereas one may recapitulate the steady-state density profiles by calibrating only the weights w_{ij} in Eq. (12) directly, the predictions of nonequilibrium conditions are more accurate if $g_{\ell i}$ are explicitly accounted for. In particular, the explicit estimation of the state degeneracies $g_{\ell i}$ is crucial in Eqs. (S16)–(S19) for the transport coefficients derived from the local Boltzmann distribution (Sec. S5).

To relate our simulations to realistic physical scales, hard-core gas can be compared to argon [2] with the molecules' diameter $\sigma = 366$ pm (Van der Waals radius 183 pm) at $T_{\text{ref}} = \mathcal{E}/k_{\text{B}} = 300$ K (Table S1). Thereby, our coarse-grained model is calibrated at room-temperature conditions. It accurately reproduces equilibrium conditions $T \lesssim 300$ K (Fig. S1) and regimes far from equilibrium (Fig. 3).

S2 Master equation

Equation (9) divided by the total number of particles N reads

$$p_I(t+\tau) = \sum_J P_{IJ}(\tau) p_J(\tau),$$

where p_I and P_{IJ} stand for the probability of finding a particle in the state I and the transition probability from the state J to I. Taking the time derivative of $\ln p_I(t+\tau)$ with respect to τ and using Eq. (10), we then

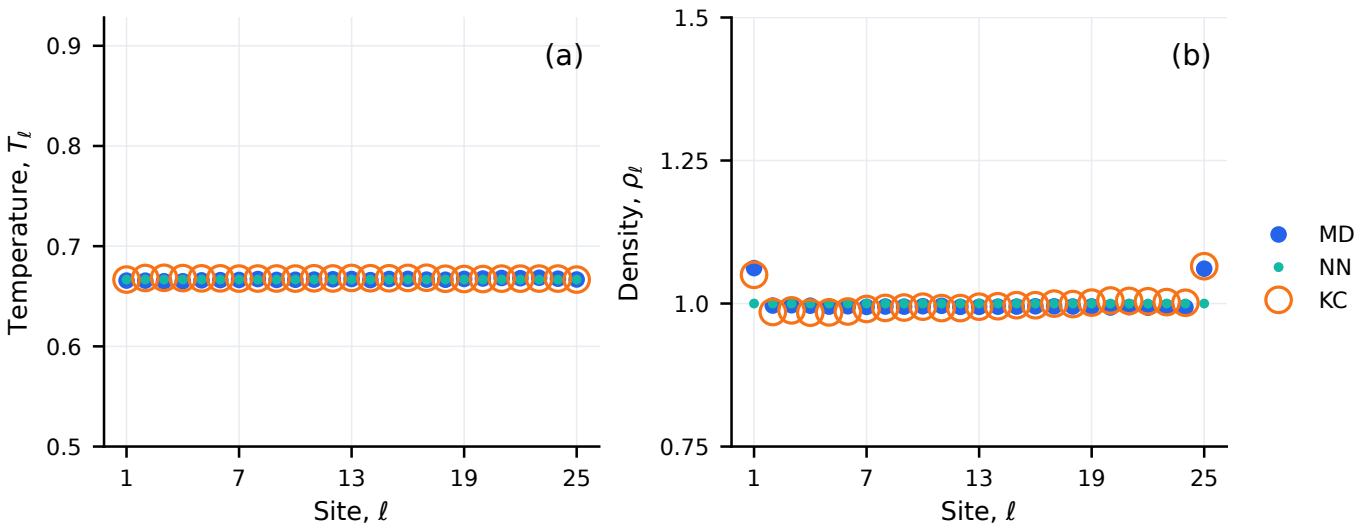


Figure S1: Equilibrium profiles of temperature (a) and density (b) for a hard-core gas confined between thermalizing walls at $\beta_{\text{left}} = \beta_{\text{right}} = 1.5$ (arb. u.). Our coarse-grained model accurately predicts histograms of temperature and density in molecular-dynamics (MD) simulations for both dynamical schemes: the kinetic connectivity (KC), calibrated at a lower value of $\beta_{\text{ref}} = 1.0$, and the nearest-neighbor transitions (NN). By construction, the NN case does not account for the minor boundary effects present at the end sites $\ell \in 1, 25$ of MC and KC simulations.

obtain

$$\frac{\partial_\tau p_I(t+\tau)}{p_I(t+\tau)} = \frac{\sum_J \partial_\tau P_{IJ}(\tau) p_J(t)}{\sum_J P_{IJ}(\tau) p_J(t)} = \frac{\sum_J K_{IJ} p_J(t)}{\sum_J P_{IJ}(\tau) p_J(t)}. \tag{S2}$$

Because $\lim_{\tau\to 0} P_{IJ}(\tau) \to \delta_{IJ}$, Eq. (S2) yields the master equation

$$\dot{p}_I(t) = \sum_J K_{IJ} p_j(t). \tag{S3}$$

S3 Reflective boundary conditions for nearest-neighbor transitions

In equilibrium, the Boltzmann distribution of the nearest-neighbor connectivity with all $g_I = g$ reads $\bar{p}_I = g \exp(-\beta\epsilon_i)/\bar{Z}(\beta)$ with the canonical partition function

$$\bar{Z}(\beta) = gL \sum_i \exp(-\beta\epsilon_i),$$

which must solve the stationary Eq. (12) at a fixed $\beta_\ell = \beta \equiv$ const. For the nearest-neighbor dynamics, $a_{IJ} = a_{\ell m}$ and $\gamma_{IJ} = \gamma_{\ell m}$, this condition implies

$$\bar{p}_I = \sum_J P_{IJ}(\beta) \bar{p}_J = \bar{p}_I \sum_m \frac{a_{\ell m}\gamma_{\ell m}}{\sum_\ell a_{\ell m}\gamma_{\ell m}}, \tag{S4}$$

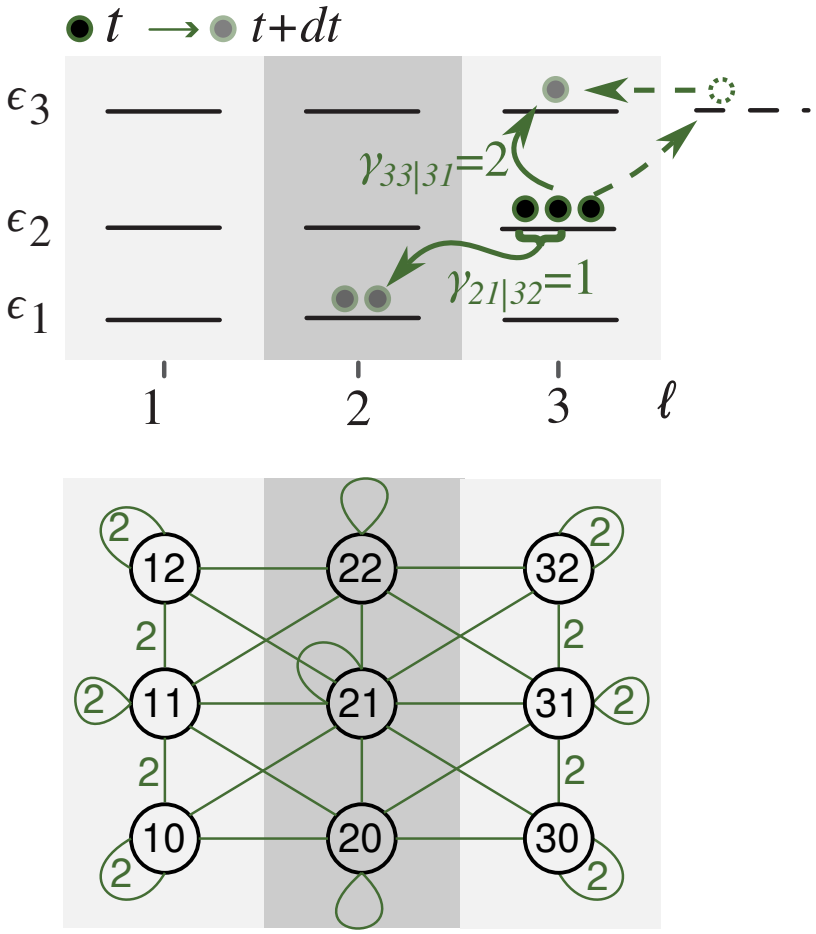


Figure S2: Boltzmann gas on a lattice of size $L = 3$ with energy levels $\epsilon_{i=1,2,3}$ and transitions $(mj) \to (\ell i)$ limited to the nearest-neighbor range $|\ell - m| \leq 1$ under reflective boundary conditions. Top: A realization of transitions $n_{21|31} = 2$ and $n_{33|31} = 1$ particles (green solid arrows). Path multiplicities emerge due to the reflective boundary conditions (green dashed arrows for the transition $n_{32|31}$). Bottom: Graph of energy states (circles with the double index) where edges denote transitions between the nearest-neighbor sites of the underlying lattice. The adjacency matrix is symmetric, $a_{\ell i|mj} = a_{mj|\ell i}$ for all these transitions (green lines). Path multiplicities $\gamma_{\ell i|mj} = 2$ enhanced due to the reflective boundary conditions are indicated by numbers next to the graph edges.

and then $\gamma_{\ell m} = \gamma_{m\ell}$, since $a_{\ell m} = a_{m\ell}$.

To ensure Eq. (S4), in the nearest-neighbor connectivity graph the reflective boundaries essentially compensate the missing transition edges for $1 \to 0$ and $L \to L+1$ at the lattice ends $\ell \in \{1, L\}$ with the multiplicities $\gamma_{1i|1j} = \gamma_{Li|Lj} = 2$. In the bulk of the system we have $\gamma_{IJ} = 1$. The left wall effectively redirects all attempts to move from the site $\ell = 1$ to the site $m = 0$ (outside of the system's confinement) back to the site $m = 1$, thus doubling the the path realizations between the energy levels $\ell = m = 1$ (Fig. S2). And likewise the transitions from $\ell = L$ to $m = L + 1$ rebound at the right end back to $m = \ell = L$.

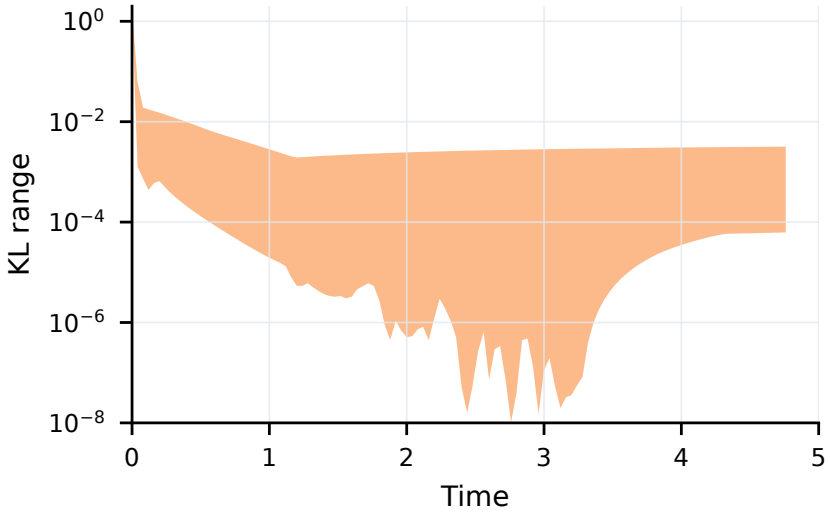


Figure S3: Range of Kullback-Leibler (KL) divergence values $[\min_\ell D^\ell_{\rm KL}, \max_\ell D^\ell_{\rm KL}]$ in the simulations of Boltzmann gas with the kinetic-connectivity dynamics at $\beta_{\rm left} = 1.1$, $\beta_{\rm right} = 1.9$ (arb. u.). The initial distribution is uniform over energy levels across all sites $\ell = 1, 2, ...L$. After the first few time steps, KL divergence drops to $\sim 10^{-3}$ and exhibits just a numerical noise close to zero.

S4 Markovian dynamics of Boltzmann gas

To solve Eq. (12) for the probability distribution $p_I(t)$ of particles over the energy levels $I = (\ell, i)$, which numerically approximates Eq. (11) by an Euler step of size Δt, we begin with a uniform distribution $p_I(0) = (LM)^{-1}$. For $\ell = 2, 3, ...L$, the initial guesses of β_ℓ are refined via Gauss-Seidel iterations [3, 4] so as to satisfy the energy constraint Eq. (5) within a given tolerance.

Solutions of both, nearest-neighbor and kinetic connectivity schemes, converge to local equilibrium within a few steps, as can be quantified by the Kullback-Leibler divergence (Fig. S3). In particular, we evaluate the time series of

$$D^\ell_{\rm KL} = D_{\rm KL}\Big(p_{i|\ell}(t) \parallel \bar{p}_i[\beta_\ell(t)]\Big) = \sum_{i=1}^{M} p_{i|\ell}(t) \ln \frac{p_{i|\ell}(t)}{\bar{p}_i\big(\beta_\ell(t)\big)},$$

where $p_{i|\ell}(t) = p_{\ell i}(t) / \sum_i p_{\ell i}(t)$ is the conditional distribution of particles at site ℓ and time t, and

$$\bar{p}_i(\beta_\ell) = \frac{g_{\ell i} \exp(-\beta_\ell \epsilon_i)}{z_\ell}$$

is the canonical distribution with the local partition function $z_\ell = \sum_{i'} g_{\ell i'} \exp(-\beta_\ell \epsilon_i)$.

S5 Derivation of thermodiffusion equation

Although thermodiffusive processes described by Eq. (1) may in general contain inhomogeneous diffusion and thermodiffusion coefficients, e.g. due to the variation of local temperature, it is typical to consider their values

fixed ignoring space-dependent corrections [5]. We derive such a description within the linear-response theory from the condition of local equilibrium, verified in our numerical solution and allowing us to represent

$$p_I(t) = p_{\ell i} = \bar{p}_i\big(\beta_\ell(t)\big)\ p_\ell(t),$$

in which $p_\ell = \sum_i p_{\ell i}$ is the fraction of N particles at the site ℓ, and $\bar{p}_i(\beta_\ell)$ is the local distribution of these particles over energy levels.

Thereby we separate $p_\ell(t)$, which is an explicit function of ℓ, from $\bar{p}_i(\beta_\ell)$ that depends on the inverse temperature β_ℓ and, therefore, implicitly also on ℓ. The following derivation then relies on the linear-response theory for a profile of

$$\beta_{\ell\pm r} = \bar{\beta} + \delta\beta_{\ell\pm r} = \bar{\beta} + r\nabla\beta\Delta x$$

with constant $\bar{\beta} = \beta_\ell$ and $\nabla\beta$ over $x_{\ell\pm r} = x_\ell \pm r\Delta x$.

By summing Eq. (12) over the energy levels i, in the bulk of a large system $L \to \infty$ we get

$$p_\ell(t+\Delta t) = \sum_m P_{\ell m} p_m(t) = P_{\ell\ell} p_\ell(t) + \sum_{r=1}^{\infty} \left\{ P_{\ell(\ell+r)} p_{\ell+r}(t) + P_{\ell(\ell-r)} p_{\ell-r}(t) \right\} \tag{S5}$$

with transition probabilities

$$P_{\ell m} = \sum_{ij} P_{\ell i|mj}(\Delta t)\bar{p}_j(\beta_m). \tag{S6}$$

Using the standard expansion of Markov chains in power series of time and space [6], up to $O(\Delta t^2)$ and $O(\Delta x^3)$ respectively, on the left-hand side (LHS) of Eq. (S5) we obtain

$$\text{LHS} = p_\ell(t) + \Delta t \partial_t p_\ell(t) + O(\Delta t^2), \tag{S7}$$

and on the right hand side

$$\begin{aligned}\text{RHS} = &\left\{ P_{\ell\ell} + \sum_r \left[P_{\ell(\ell+r)} + P_{\ell(\ell-r)} \right] \right\} p_\ell(t) \\ &+ \sum_{r=1}^{\infty} \left\{ \left[P_{\ell(\ell+r)} - P_{\ell(\ell-r)} \right] r\Delta x\ \partial_x p_\ell + \left[P_{\ell(\ell+r)} + P_{\ell(\ell-r)} \right] \frac{(r\Delta x)^2}{2} \partial_x^2 p_\ell \right\} + O(\Delta x^3) = \\ &P_\ell p_\ell + \partial_x p_\ell \Sigma_T + \partial_x^2 p_\ell \Sigma_D + O(\Delta x^3),\end{aligned} \tag{S8}$$

in which we introduced kinetic rate $P_\ell = \sum_m P_{\ell m}$ and, respectively, a thermal and diffusive moments

$$\Sigma_T(\ell) = \Delta x \sum_r r \left[P_{\ell(\ell+r)} - P_{\ell(\ell-r)} \right] = \Delta x \sum_m (m-\ell) P_{\ell m}, \quad , \tag{S9}$$

$$\Sigma_D(\ell) = \frac{\Delta x^2}{2} \sum_r r^2 \left[P_{\ell(\ell+r)} + P_{\ell(\ell-r)} \right] = \frac{\Delta x^2}{2} \sum_m (m-\ell)^2 P_{\ell m}. \tag{S10}$$

Then, using the expansion up to $O(\nabla\beta^2)$, we simplify

$$\begin{aligned}P_\ell = P_{\ell\ell} + \sum_{r>1} \Big(P_{\ell(\ell+r)} + P_{\ell(\ell-r)} \Big) &= P_{\ell\ell}(\bar{\beta}) + \sum_{r>1} \Big(P_{\ell(\ell+r)}(\bar{\beta}) + P_{\ell(\ell-r)}(\bar{\beta}) \Big) + O(\nabla\beta^2) \\ &\simeq P_{\ell\ell}(\bar{\beta}) + \sum_{r>1} \Big(P_{(\ell+r)\ell}(\bar{\beta}) + P_{(\ell-r)\ell}(\bar{\beta}) \Big) + O(\nabla\beta^2) = 1 + O(\nabla\beta^2),\end{aligned} \tag{S11}$$

$$r\Delta x[P_{\ell(\ell+r)} - P_{\ell(\ell-r)}] = r\Delta x\left[P_{\ell(\ell+r)}(\bar\beta) - P_{\ell(\ell-r)}(\bar\beta) + \delta\beta_r\partial_\beta P_{\ell(\ell+r)}(\bar\beta) + \delta\beta_r\partial_\beta P_{\ell(\ell-r)}(\bar\beta)\right] + O(\delta\beta_r^2)$$
$$\simeq (r\Delta x)^2\nabla\beta\left[\partial_\beta P_{\ell,\ell+r} + \partial_\beta P_{\ell,\ell-r}\right] + O(\nabla\beta^2), \quad \text{(S12)}$$

which neglect the boundary artifacts in a large system by assuming a homogeneity of transition probability $P_{\ell(\ell+r)}(\bar\beta) = P_{\ell(\ell-r)}(\bar\beta)$ in equilibrium at $\bar T = (k_\mathrm{B}\bar\beta)^{-1}$. Thanks to Eq. (S12) we simplify (S9) to

$$\Sigma_T = \nabla\beta\Delta x^2\sum_m (m-\ell)^2\partial_\beta P_{\ell m}(\bar\beta) + O(\nabla\beta^2). \quad \text{(S13)}$$

and then we also have

$$\Sigma_T\partial_x p_\ell = \partial_x(\Sigma_T p_\ell) - \partial_x\Sigma_T p_\ell = \partial_x(\Sigma_T p_\ell) + O(\nabla\beta^2), \quad \text{(S14)}$$
$$\Sigma_D\partial_x^2 p_\ell = \partial_x(\Sigma_D\partial_x^2 p_\ell) - \partial_x\Sigma_D\partial_x p_\ell$$
$$= \partial_x(\Sigma_D\partial_x^2 p_\ell) - \partial_x(\partial_x\Sigma_D p_\ell) + \partial_x^2\Sigma_T p_\ell = \partial_x(\Sigma_D\partial_x^2 p_\ell) - \partial_x(\partial_x\Sigma_D p_\ell) + O(\nabla\beta^2). \quad \text{(S15)}$$

Note that $\partial_\beta\Sigma_D = O(\Delta x^3)$, but

$$\partial_x\Sigma_D = \frac{\nabla\beta\Delta x^2}{2}\sum_m (m-\ell)^2\partial_\beta P_{\ell m}(\bar\beta) = O(\nabla\beta, \Delta x^2),$$

which we must account for.

Since $\rho \propto \lim_{\Delta x\to 0} p/\Delta x$ and $\nabla\beta = -\nabla T/\bar T^2 + O(\nabla\beta^2)$, by combining Eqs. (S5)–(S15) and taking the continuum limit $\Delta x \to 0$, $\Delta t \to 0$, in the linear regime we obtain

$$\partial_t\rho = -\partial_x j_m, \qquad j_m = -\Big(D_T\nabla T\rho + D\partial_x\rho\Big), \quad \text{(S16)}$$

in which we identify the coefficients

$$D_T(\bar T) = -\frac{\nabla T}{\bar T^2}\lim_{\substack{\Delta x\to 0\\ \Delta t\to 0}}\frac{1}{\nabla\beta\Delta t}\Big(\Sigma_T - \partial_x\Sigma_D\Big) = -\frac{\nabla T}{\bar T^2}\lim_{\substack{\Delta x\to 0\\ \Delta t\to 0}}\frac{\Delta x^2}{2\Delta t}\sum_m (m-\ell)^2\partial_\beta P_{\ell m}\Big((k_\mathrm{B}\bar T)^{-1}\Big), \quad \text{(S17)}$$

$$D(\bar T) = \lim_{\substack{\Delta x\to 0\\ \Delta t\to 0}}\frac{1}{\Delta t}\Sigma_D = \lim_{\substack{\Delta x\to 0\\ \Delta t\to 0}}\frac{\Delta x^2}{2\Delta t}\sum_m (\ell-m)^2 P_{\ell m}\Big((k_\mathrm{B}\bar T)^{-1}\Big). \quad \text{(S18)}$$

The linear-response formulas Eqs. (S17) and (S18) can be evaluated also for finite Δx and Δt to estimate the Soret coefficient in the system bulk at $\bar\ell = \lfloor (L+1)/2\rfloor$:

$$S_T = D_T(\bar\ell, \bar\beta)/D(\bar\ell, \bar\beta). \quad \text{(S19)}$$

References

[1] M. N. Bannerman, R. Sargant, and L. Lue. DynamO: a free $\mathcal{O}(N)$ general event-driven molecular dynamics simulator. *Journal of Computational Chemistry* 32.15 (2011), pp. 3329–3338. ISSN: 1096-987X. DOI: 10.1002/jcc.21915.

[2] *WebElements: the periodic table on the web*. https://webelements.com/. Accessed: 2026-06-04.

[3] J.M. Ortega and W.C. Rheinboldt. "General iterative methods". *Iterative Solution of Nonlinear Equations in Several Variables*. Elsevier, 1970, pp. 181–239. ISBN: 9780125285506. DOI: 10.1016/b978-0-12-528550-6.50018-1.

[4] Code supporting the findings of this paper is available at https://git.embl.org/rbelouso/microtherm.

[5] Roman Belousov, Ali Hassanali, and Édgar Roldán. Statistical physics of inhomogeneous transport: Unification of diffusion laws and inference from first-passage statistics. *Physical Review E* 106.1 (2022). ISSN: 2470-0053. DOI: 10.1103/physreve.106.014103.

[6] R. Belousov et al. Microcanonical ensemble out of equilibrium. *Physical Review E* 113.1 (2026). ISSN: 2470-0053. DOI: 10.1103/r1cg-tpdb.